

American Option Pricing Using a Trinomial Tree

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1 Introduction

This report explains the methodology and mathematical foundations behind the trinomial tree method used to price American options. Special attention is given to the backward induction algorithm implemented in the code, and the behavior of the early exercise boundary for put and call options. The main question we attempt to answer is : "When is it optimal to exercise an American Option?". The answer defines the Early Exercise Boundary.

2 Methodology

2.1 Backward Induction in the Trinomial Tree

The core of the trinomial tree method for pricing American options is the backward induction loop:

```
1 for step in range(n, 0, -1):  
2     option_values = discount * (  
3         pu * option_values[2:] +  
4         pm * option_values[1:-1] +  
5         pd * option_values[:-2]  
6     )
```

This calculates option values at time $t = \Delta t \times \text{step}$, moving backward from maturity T to the present.

The option lifetime T is divided into n steps of length $\Delta t = \frac{T}{n}$. At each step, the asset price moves up, down, or remains the same with risk-neutral probabilities p_u, p_m, p_d . The asset price at node j and time t is:

$$S_{t,j} = S_0 \exp(vj), \quad v = \lambda \sigma \sqrt{\Delta t}, \quad j = -t, \dots, t.$$

The backward induction formula for the option value at node (t, j) is:

$$V_{t,j} = e^{-r\Delta t} (p_u V_{t+\Delta t, j+1} + p_m V_{t+\Delta t, j} + p_d V_{t+\Delta t, j-1}),$$

with risk-free rate r .

2.2 Vectorized Computation and Array Slicing

Representing option values at time t as:

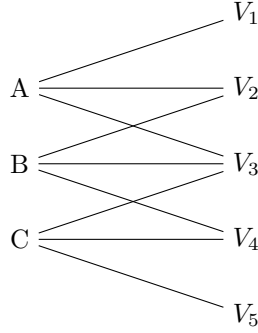
$$\text{option_values} = [V_1, V_2, V_3, V_4, V_5],$$

the backward induction step computes:

$$\text{discount} \times (p_u \times \text{option_values}[2:] + p_m \times \text{option_values}[1:-1] + p_d \times \text{option_values}[:-2]),$$

where

$$\begin{aligned} \text{option_values}[:-2] &= [V_1, V_2, V_3], \\ \text{option_values}[1:-1] &= [V_2, V_3, V_4], \\ \text{option_values}[2:] &= [V_3, V_4, V_5]. \end{aligned}$$



2.3 Matrix Representation of Backward Induction

The backward induction can be expressed as:

$$V_t = e^{-r\Delta t} \cdot P \cdot V_{t+\Delta t},$$

where P is the transition matrix:

$$P = \begin{bmatrix} p_d & p_m & p_u & 0 & 0 \\ 0 & p_d & p_m & p_u & 0 \\ 0 & 0 & p_d & p_m & p_u \end{bmatrix},$$

with each row corresponding to a node at time t and columns to nodes at $t + \Delta t$.

For American options, early exercise is included by:

$$V_t = \max(V_t, I_t),$$

where I_t is the intrinsic value vector.

2.4 Illustrative Example

Suppose at time $t + \Delta t$, the option values vector is:

$$V_{t+\Delta t} = \begin{bmatrix} 10 \\ 12 \\ 14 \\ 16 \\ 18 \end{bmatrix},$$

and the transition probabilities are:

$$p_d = 0.2, \quad p_m = 0.5, \quad p_u = 0.3.$$

Multiplying P by $V_{t+\Delta t}$ yields:

$$V_t = e^{-r\Delta t} \times \begin{bmatrix} 0.2 \times 10 + 0.5 \times 12 + 0.3 \times 14 \\ 0.2 \times 12 + 0.5 \times 14 + 0.3 \times 16 \\ 0.2 \times 14 + 0.5 \times 16 + 0.3 \times 18 \end{bmatrix}.$$

This matches the vectorized code, where slices correspond to:

$$\begin{aligned} \text{option_values}[:-2] &= [10, 12, 14], \\ \text{option_values}[1:-1] &= [12, 14, 16], \\ \text{option_values}[2:] &= [14, 16, 18]. \end{aligned}$$

Thus, the code computes simultaneously:

$$V_t = e^{-r\Delta t} \times (p_d \times [10, 12, 14] + p_m \times [12, 14, 16] + p_u \times [14, 16, 18]).$$

3 Early Exercise Boundary Behavior

3.1 Definition of the Early Exercise Boundary $B(t)$

The early exercise boundary $B(t)$ is a function of time that defines the critical underlying asset price level below or above which it is optimal to exercise the American option early. Specifically:

$B(t)$ = the stock price at time t where the option holder is indifferent between exercising and holding.

3.2 General Behavior

American Put Options For American puts, the early exercise boundary converges near the strike price K as maturity approaches. Exercising early is beneficial when the underlying price S_t falls sufficiently below K , allowing the holder to realize intrinsic value immediately.

$$B(t) \approx K \quad \text{as } t \rightarrow T$$

American Call Options with Dividends For American calls on dividend-paying stocks, the early exercise boundary converges to a value **above** the strike price K , as early exercise is optimal only when the stock price is sufficiently higher than K to compensate for the loss of time value and to capture dividends.

$$B(t) > K \quad \text{as } t \rightarrow T$$

Without dividends, early exercise of call options is generally suboptimal, and the boundary may not exist.

Effect of Volatility Higher volatility increases the option's time value by raising the potential for favorable future price movements. Since early exercise forfeits this time value, greater volatility generally reduces the incentive to exercise early. As a result, the early exercise boundary shifts to delay exercise—moving higher for calls and lower for puts.

3.3 Validity and Limitations of Observations

The observed boundary behaviors hold broadly under typical market conditions and standard modeling assumptions, but:

- Extreme parameter values (interest rates, volatility, dividend yields) can alter boundary behavior.
- The results assume the Black-Scholes framework extended with a trinomial tree; real-world factors like transaction costs may change early exercise strategies.

3.4 Early Exercise Boundary Figures and Analysis

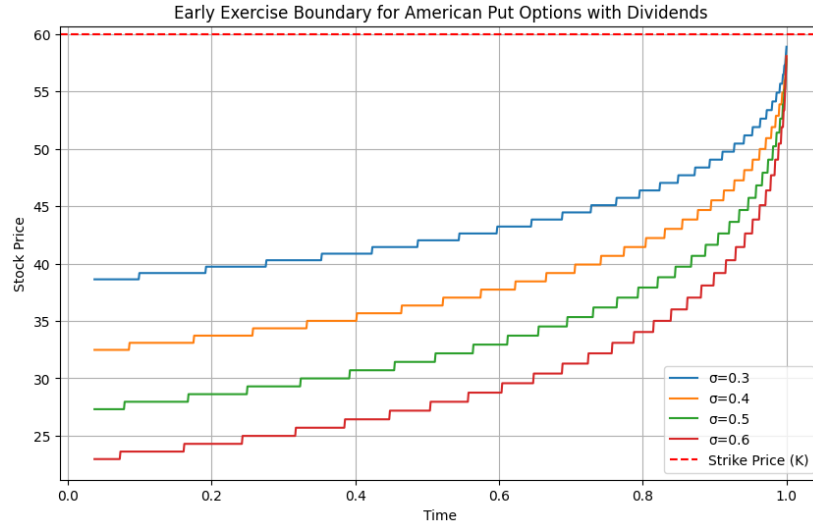


Figure 1: Early Exercise Boundary for American Put Options with Dividends under various volatilities.

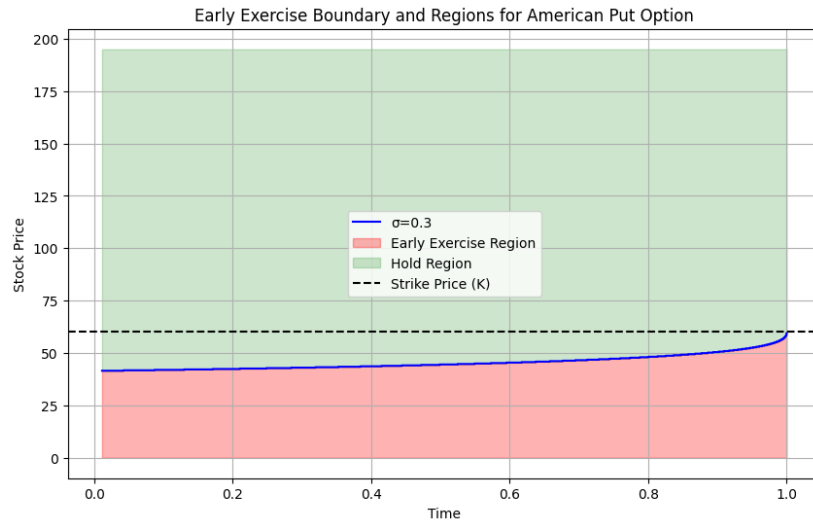


Figure 2: Early Exercise Boundary for American Put Options with Dividends with shaded regions.

Analysis: This plot shows the early exercise boundaries $B(t)$ for American put options with dividends under different volatility levels ($\sigma = 0.3, 0.4, 0.5, 0.6$). The stock price S_t at which early exercise becomes optimal is plotted against time t .

The early exercise boundary increases as time approaches maturity, converging towards the strike price K (dashed red line). Higher volatility corresponds to a lower early exercise boundary, indicating that with increased uncertainty, it is optimal to wait longer before exercising.

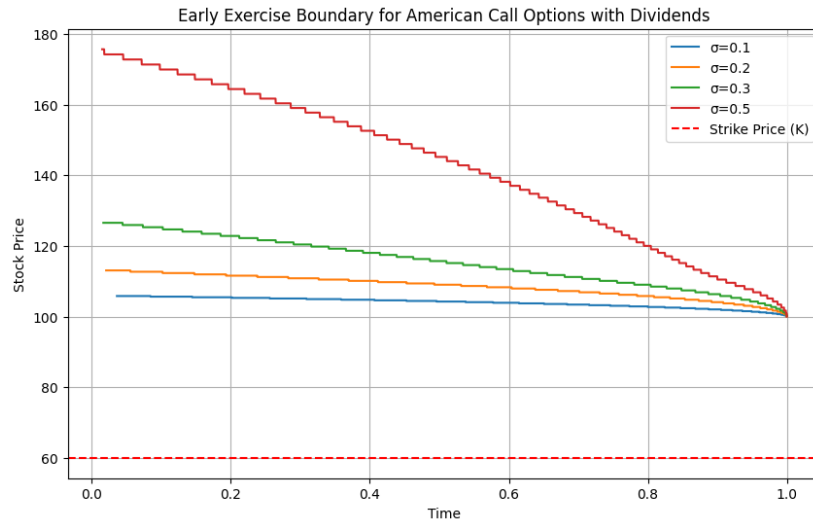


Figure 3: Early Exercise Boundary for American Call Options with Dividends under various volatilities.

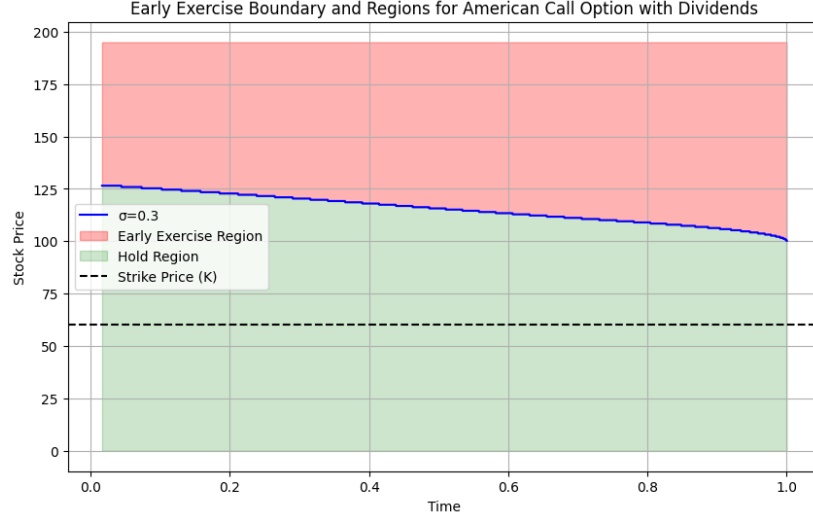


Figure 4: Early Exercise Boundary for American Call Options with Dividends with shaded regions.

Analysis: This plot illustrates the early exercise boundaries $B(t)$ for American call options with dividends across several volatility levels ($\sigma = 0.1, 0.2, 0.3, 0.5$).

Unlike puts, the early exercise boundary for calls lies above the strike price K (dashed red line). Higher volatility tends to raise the boundary, reflecting greater option time value and less incentive to exercise early. The impact of dividends is crucial here, as early exercise is only optimal to capture dividend payments.

3.5 Key Observation

For American Call Options, we know that if we exercise early we get to enjoy the dividend payment. $B(t)$ for the American call option approaches the strike price from above. We early exercise if our point is above this boundary; otherwise, we hold if it's between the boundary and the strike. The reason why there is a small gap where the boundary doesn't converge exactly to the strike is because exercising exactly at the strike yields zero intrinsic value and forfeits remaining time value, so the option holder prefers to wait unless the stock price is sufficiently above the strike to compensate for this loss and capture dividends.

For American put options, the boundary $B(t)$ converges directly to K from below as maturity approaches. Since exercising the put below K provides immediate positive payoff, and time value diminishes near expiry, early exercise is optimal whenever the stock price is less than K .

In summary, calls tend to delay exercise just above K to preserve time value, while puts exercise promptly below K near maturity.

4 Code for Trinomial Tree for American Put Options with Dividends

```

1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 def trinomial_tree_american_put_boundary_dividend(S0, K, r, q,
5           sigma, T, n, lam):
6     dt = T / n
7     v = lam * sigma * np.sqrt(dt)
8
9     pu = 1 / (2 * lam**2) + (r - q - 0.5 * sigma**2) * np.sqrt(dt)
10    / (2 * lam * sigma)
11    pd = 1 / (2 * lam**2) - (r - q - 0.5 * sigma**2) * np.sqrt(dt)
12    / (2 * lam * sigma)
13    pm = 1 - 1 / (lam**2)
14
15    discount = np.exp(-r * dt)
16
17    ST = S0 * np.exp(v * np.arange(-n, n + 1))
18    option_values = np.maximum(K - ST, 0)
19
20    boundary = [None] * n
21
22    for step in range(n, 0, -1):
23        option_values = discount * (
24            pu * option_values[2:] +
25            pm * option_values[1:-1] +
26            pd * option_values[:-2]
27        )
28        ST = S0 * np.exp(v * np.arange(-step + 1, step))
29        intrinsic_values = np.maximum(K - ST, 0)
30
31        exercise = intrinsic_values > option_values
32        if np.any(exercise):
33            boundary[step-1] = max(ST[exercise])
34        else:
35            boundary[step-1] = None
36
37        option_values = np.maximum(option_values, intrinsic_values)
38
39    times = np.linspace(dt, T, n)
40    return times, boundary
41
42 # Parameters
43 S0 = 65
44 K = 60
45 r = 0.05
46 q = 0.03 # Dividend yield
47 T = 1
48 n = 2**11
49 lam = 1.5
50
51 sigmas = [0.3, 0.4, 0.5, 0.6]
52 plt.figure(figsize=(10,6))

```



```

51 for sigma in sigmas:
52     times, boundary = trinomial_tree_american_put_boundary_dividend
        (S0, K, r, q, sigma, T, n, lam)
53     filtered_times = [t for t, b in zip(times, boundary) if b is
        not None]
54     filtered_boundary = [b for b in boundary if b is not None]
55     plt.plot(filtered_times, filtered_boundary, label=f'\u03c3={
        sigma}')
56
57 plt.axhline(y=K, color='red', linestyle='--', label='Strike Price (
    K)')
58 plt.xlabel('Time')
59 plt.ylabel('Stock Price')
60 plt.title('Early Exercise Boundary for American Put Options with
    Dividends')
61 plt.legend()
62 plt.grid(True)
63 plt.show()

```

Listing 1: Trinomial Tree for Early Exercise Boundary of American Put Options with Dividends